S3 Table: Classification of the histopathological findings in COVID-19 cases

Organ	Taxonomy	Histopathological findings	Prevalence
Lung	Diffuse Alveolar Damage	• DAD any phase (acute, organizing, fibrotic) • Hyaline membranes • Alveolar fibrin plugs • Loss of pneumocytes • Reactive pneumocytes and syncytial cells • Vesicular nuclei with prominent nucleoli • Type 2 pneumocyte hyperplasia • Microthrombi • Increased intra alveolar megakaryocytes	315
	Coagulopathy associated abnormalities	• Microthrombi	173
	Endothelial injury	• Endotheliitis • Endothelial necrosis	61
	Thromboembolic disease	• Medium/large vessel thromboembolism • Pulmonary infarction	47
	Vasculitis	• Vasculitis • Capillaritis • Diffuse alveolar hemorrhage • Alveolar hemorrhage • Pulmonary hemorrhage	10
	Possible cytopathic effect encompasses	• Atypical polymorphous pneumocytes • Apoptotic bodies • Intranuclear inclusions	21
	Serositis	• Acute pleuritis	1
	Non-specific lung scarring	• Proliferation myofibroblastic • Organizing pneumonia pattern • Interstitial fibrosis • Lymphocyte infiltration • Prominent lymphoid infiltrates • Lymphocytes CD4/CD8 • Interstitial T-cell reaction • Angiogenesis/Capillary proliferation • Squamous metaplasia	103
	Changes due to hemodynamic compromise	• Pulmonary capillary congestion • Intra alveolar edema	27
	Complications	• Bacterial pneumonia	94
		• Hemophagocytosis	1
	Pre-existent or non-significant conditions	• Emphysema • Amyloidosis of pulmonary vessels • Mononuclear macrophage	29
Upper airways inflammation	Inflammation	• Pharyngeal hyperemia • Tracheitis • Edema bronchial mucosa	33
Liver	Possibly pre-existent parenchymal abnormalities encompass	• Steatosis = fatty changes = NASH • Cholestasis, ductal proliferation	93
	Endothelial injury	• Endotheliitis	1
	Thromboembolic disease	• Venous outflow obstruction • Post vein phlebosclerosis (portal ?) • Thrombosis	93

	Possible cytopathic effect encompasses	• Apoptotic hepatocytes	2
		• Hepatitis: Portal inflammation, periportal and lobular T-cell reaction, Lymphocytic infiltrate, Lobular inflammation, Kupffer cell proliferation	50
	Abnormalities secondary to hemodynamic compromise	• Hepatomegaly • Edema/congestion • Shock liver (common perimortem terminal event) • Aberrant portal vessels • Periportal abnormal vessels	114
	Non-specific scarring	Fibrosis NOS	60
	Pre-existent or non-significant conditions :	Cirrhosis, Gall bladder – enlarged, Liver cyst	10
CARDIOVASCULAR	Endothelial injury	• Endotheliitis	8
	Thromboembolic disease	• Thrombosis	45
	Possible cytopathic effect	• Cardiomyocyte necrosis • Apoptotic bodies • Myocarditis (24)- Interstitial infiltration, Lymphocytic and Eosinophilic myocarditis	37
	Serositis	• Epicarditis • Epicardial inflammation • Pericardial edema • Pericardial effusion	4
	Complications	• Cardiomyopathy (if not pre-existent) • Hemophagocytosis	5
	Pre-existent cardiovascular disease	• Myocardial/myocyte hypertrophy • Atherosclerosis • Ischemic cardiomyopathy • MI • Myocardial/interstitial fibrosis • Cardiomegaly	115
	Pre-existent or non-significant conditions	• Amyloidosis	8
KIDNEY/GENITOURINARY	Endothelial injury	• Endotheliitis	1
	Thromboembolic disease	• Artery thrombus • Thromboemboli	2
	Possible viral effect	• Glomerulopathies (if not pre-existent) Includes FSGS, MNS, mesangial expansion	9
	Abnormalities secondary to hemodynamic compromise	• Congestion • Shock kidney • Congestion/medullary edema • Acute tubular injury	147
	Coagulopathy associated abnormalities	• DIC • Capillary thrombus • Segmental fibrin thrombus	55
	Non-specific scarring	• Granular surface/scarring • Cortical fibrosis	24
	Pre-existent vascular disease	• Arterionephrosclerosis • Benign nephrosclerosis	76
	Pre-existent or non-significant conditions	• Cysts • Testicular atrophy • Amyloidosis	47

		• Pigment casts • Glomerulopathies (if present before infection) (FSGS, MNS, mesangial expansion) • Prostate thrombosis • Benign prostatic hyperplasia • Hypospermatogenesis • Mild peritubular hyalinization	
Central Nervous System	Abnormalities secondary to hemodynamic compromise	• Cerebral edema/congestion • Hypoxic injury • Anoxic injury	34
	Coagulopathy associated abnormalities	• Hemorrhage • Microthrombi	20
	Thromboembolic disease	• Infarcts (if recent)	9
	Inflammation (Possible cytopathic effect)	• Leptomeningeal inflammation • Perivascular inflammation • Encephalitis: 5 • Lymphocytic meningitis	9
	Pre-existent vascular disease:	• Cerebral chronic infarcts • Atherosclerosis	18
	Pre-existent or non-significant conditions	• Subdural hematoma • Cerebral cortex atrophy • Pale substantia nigra • Focal spongiosis • Lewy body disease • Alzheimer's disease • Microglial nodule • Neuronal cell loss • Axonal degeneration • Residual anaplastic astrocytoma • Cytolysis	35
GASTROINTESTINAL	Endothelial injury	• Endotheliitis	2
	Coagulopathy associated abnormalities	• Hemorrhage	3
	Abnormalities secondary to hemodynamic compromise or thromboembolic disease	• Mesenteric necrosis • Ischemic enterocolitis • Pancreatitis	15
	Possible cytopathic effect	• Apoptotic bodies	2
	Pre-existent or non-significant conditions	• Esophagitis • Diverticulosis • Chronic gastritis	2
HEMATOLYMPHOID	Possible cytopathic effect	• Increased plasma blasts • Apoptosis	15
	Abnormalities secondary to hemodynamic compromise	• Congestion • Infarction	7
	Thromboembolic disease	• Arteriolar thrombosis	1
	Non-specific changes secondary to systemic infection	• Decreased myelopoiesis • Lymphadenopathy • Splenomegaly • Bone marrow hyperplasia • Left-shifted myelopoiesis • Left-shifted granulopoiesis • Splenitis • Plasma cells in spleen	41
	Changes likely due to steroid therapy	• Lymphoid depletion • Reduced white pulp	29
	Complications	• Hemophagocytosis	21

SKIN	Endothelial injury	• Endotheliitis	1
	Coagulopathy associated abnormalities	• Petechiae • Dermal necrosis • Dermo-hypodermal/superficial thrombi • Deep dermis thrombi	12
	Coagulopathy associated abnormalities	• Thrombogenic vasculopathy	1
	Vasculitis	• Livedo • Purpura • Subcutaneous lymphocytic vasculitis • Lymphocytic infiltration of vessels	11
	Possible cytopathic effect	• Intranuclear viral inclusions: 1 • Multinucleated cells: 1 • Intraepidermal vesicle • Dyskeratosis • Necrotic keratinocytes	7
	Non-specific findings associated with skin inflammation (dermatitis)	• Exocytosis • Spongiosis • Acantholysis • Vesicles • Suprabasal clefts • Dermal edema • Vascular ectasia • Dermal mucin • Dermal eosinophils • Superficial dermal inflammatory infiltrates • Perivascular deep inflammation • Peri-eccrine inflammation • Lymphocytic panniculitis • Parakeratosis	15
PLACENTA	Possibly associated with coagulopathy	• Infarcts • Fibrin deposition • Villous agglutination	11
	Inflammation	• Inflammatory infiltrates • Funisitis	1
	Common non-specific changes of varied etiology (e.g. hypertension, diabetes, smoking etc.)	• accelerated villous maturation • Hypertrophy of membrane arterioles • Decidual arteriopathy • Maternal vascular malperfusion • Necrosis of maternal vessels	15
	Pre-existent/unrelated findings	• Chorionic hemangioma	1